

# Brain-HIVE Retrieval Reproduction

Reproducing the B32 versus B32+VAE ablation on THINGS-EEG

|            |                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Student    | Yuhui Zheng                                                                                                                           |
| Paper      | Learning Brain Representation with Hierarchical Visual Embeddings (ICLR 2026)                                                         |
| Target     | Table 3: intra-subject 200-way zero-shot retrieval                                                                                    |
| Repository | <a href="https://github.com/HUI-HUI-lab/brainhive-retrieval-reproduction">github.com/HUI-HUI-lab/brainhive-retrieval-reproduction</a> |
| Run date   | 16 July 2026                                                                                                                          |

**Outcome: adding low-level VAE features raised the reproduced 10-subject mean from 21.6% to 50.2% Top-1 and from 53.1% to 82.6% Top-5. The improvement occurred for both metrics in all ten subjects, so the paper's central directional conclusion is supported.**

## Scope

This report reproduces one representative result rather than the full paper. Reconstruction, diffusion-prior training, inter-subject retrieval, and other visual-encoder combinations are out of scope.

## 1. Paper and reproduction target

Zheng et al. propose aligning brain signals with hierarchical visual representations from encoders that capture complementary semantic and low-level information. The selected target is the Table 3 THINGS-EEG intra-subject ablation comparing CLIP ViT-B/32 (B32) against B32+VAE. The paper reports 52.2%/83.3% Top-1/Top-5 for B32 and 73.6%/94.3% for B32+VAE, averaged over ten subjects.

### Reproduction question

Does adding the VAE branch produce a clear and consistent retrieval improvement under a comparable released-code workflow, even if exact scores differ?

## 2. Experimental settings

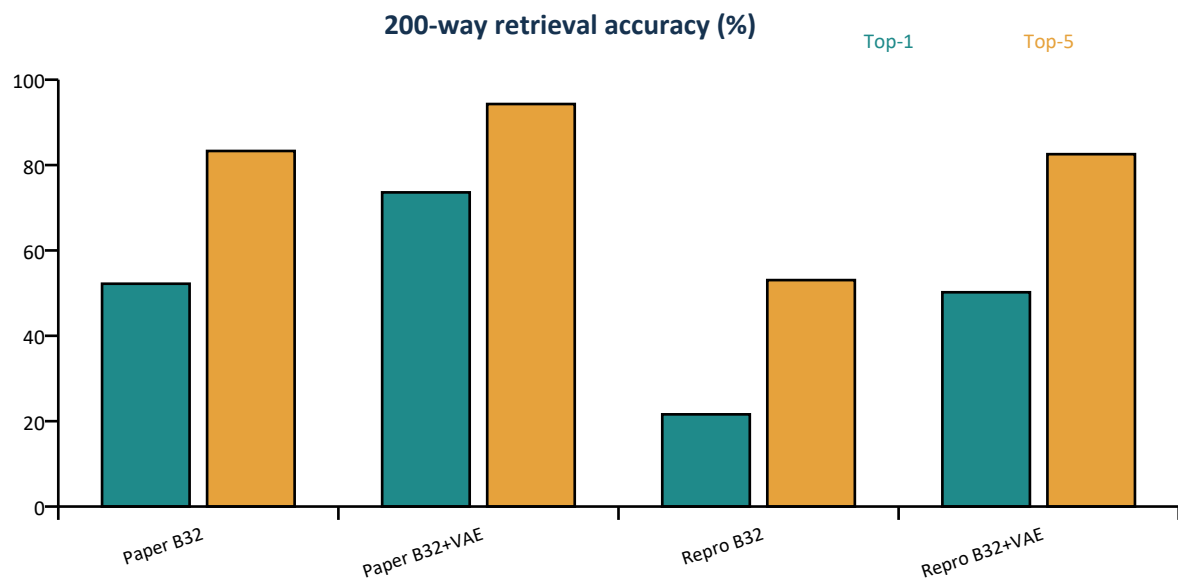
| Component | Paper                                     | Reproduction                                  |
|-----------|-------------------------------------------|-----------------------------------------------|
| Task      | THINGS-EEG intra-subject, 200-way         | Same, all 10 subjects                         |
| EEG       | 17 O+P channels; repeated trials averaged | Same; validated 17 x 250 tensors              |
| Targets   | B32; B32+VAE                              | Cached B32 512-D; VAE 1024-D                  |
| Training  | 25 epochs; AdamW; LR 5e-4; batch 1024     | Same; one seed (2025); BF16                   |
| Warmup    | 10 optimization steps                     | 10 steps (released default was 0)             |
| Metric    | Top-1 / Top-5 over 200 candidates         | Same; evaluation batch 200                    |
| Hardware  | Not a comparison variable                 | 1 x RTX 5090 32 GB                            |
| Software  | Released Brain-HIVE code                  | Python 3.12; Torch 2.9.1; Transformers 4.57.3 |

### Data validation

Each subject contained 16,540 averaged training examples and 200 averaged test examples. A CPU preflight check verified EEG shapes, embedding dimensions, and image-ID alignment before any GPU run. Public EEG files already present in a trusted read-only server cache were linked into the documented layout; the authors' 75 MB visual-embedding subset was downloaded separately.

### 3. Results

| Source / setting       | Top-1 (%) | Top-5 (%) | VAE gain (pp) |
|------------------------|-----------|-----------|---------------|
| Paper - B32            | 52.2      | 83.3      | -             |
| Paper - B32+VAE        | 73.6      | 94.3      | +21.4 / +11.0 |
| Reproduction - B32     | 21.6      | 53.1      | -             |
| Reproduction - B32+VAE | 50.2      | 82.6      | +28.6 / +29.5 |



#### Per-subject reproduced accuracy (%)

| Subj | B32 T1 | B32 T5 | B32+VAE T1 | B32+VAE T5 | T1 gain | T5 gain |
|------|--------|--------|------------|------------|---------|---------|
| 1    | 19.0   | 57.0   | 47.0       | 81.0       | +28.0   | +24.0   |
| 2    | 21.0   | 53.5   | 45.5       | 81.0       | +24.5   | +27.5   |
| 3    | 22.5   | 48.0   | 50.0       | 86.5       | +27.5   | +38.5   |
| 4    | 18.0   | 48.5   | 35.5       | 66.5       | +17.5   | +18.0   |
| 5    | 14.0   | 39.5   | 40.5       | 80.0       | +26.5   | +40.5   |
| 6    | 22.0   | 50.5   | 57.0       | 85.0       | +35.0   | +34.5   |
| 7    | 25.0   | 55.0   | 48.0       | 82.5       | +23.0   | +27.5   |
| 8    | 26.0   | 65.5   | 62.5       | 89.0       | +36.5   | +23.5   |
| 9    | 18.5   | 46.5   | 55.0       | 85.0       | +36.5   | +38.5   |
| 10   | 30.0   | 66.5   | 61.0       | 89.0       | +31.0   | +22.5   |

## 4. Interpretation

The paper's main conclusion is approximately supported. B32+VAE outperformed B32 for Top-1 and Top-5 for every subject, and the reproduced mean gains were large. This is stronger evidence than a single subject or isolated run because the trend is consistent across all ten paired comparisons.

### Why are absolute scores lower?

The reproduced B32 scores are 30.6/30.3 points below Table 3, while B32+VAE is 23.4/11.8 points lower. The released repository has evolved since the reported experiment and contains configuration inconsistencies: the public intra-subject script labels an RN50 run while selecting SynCLR, and its YAML default uses zero warmup although the paper specifies ten steps. This reproduction used explicit B32 configurations and the paper's warmup. Other plausible contributors are one random seed, BF16 arithmetic, library-version drift, and differences between the exact internal paper snapshot and the released commit.

### What can still be concluded?

Exact numerical replication is not achieved, but the causal comparison requested by the assignment is clear: holding subject, seed, optimizer, batch size, epochs, channels, and evaluation fixed, adding VAE features substantially improves retrieval. The improvement is therefore robust within the reproduced workflow.

## 5. AI-assisted workflow

Codex was used for repository inspection, experiment planning, code changes, SSH diagnostics, dependency debugging, data validation, job monitoring, aggregation, documentation, and PDF generation. GitHub CLI and standard SSH/Git commands were used for synchronization and server execution. Human decisions selected the retrieval ablation, approved server access, and authorized experiment execution.

### Representative prompts and debugging actions

| Purpose   | Representative prompt/action                                                       |
|-----------|------------------------------------------------------------------------------------|
| Planning  | Inspect Brain-HIVE and design a minimal retrieval reproduction for the assignment. |
| Server    | Check server resources, synchronize with GitHub, and run safely on an idle GPU.    |
| Data      | Validate EEG shapes, embedding dimensions, and image IDs before using a GPU.       |
| Debugging | Diagnose the smoke-test traceback without modifying shared system software.        |
| Analysis  | Compare B32 with B32+VAE and evaluate whether the paper's trend is supported.      |

## 6. Reproducibility, time, and cost

### Commands

```
python reproduction/scripts/validate_data.py --data-root DATA --subjects 1
bash reproduction/scripts/run_matrix.sh DATA RESULTS 1,2,3,4,5,6,7,8,9,10 2025
bash reproduction/scripts/aggregate.sh RESULTS
```

| Resource                  | Recorded / estimated value                                        |
|---------------------------|-------------------------------------------------------------------|
| Formal experiment runtime | 1,681 s summed across 20 runs (28 min 1 s)                        |
| Mean runtime              | B32: 74.9 s/subject; B32+VAE: 93.2 s/subject                      |
| Matrix wall-clock window  | Approximately 25 minutes for Subjects 2-10                        |
| AI-assisted working time  | Approximately 2 hours active work (estimate)                      |
| AI token usage            | Approximately 150k tokens (rough estimate; UI export unavailable) |
| Incremental AI cost       | Not separately metered; Codex subscription/account usage          |
| GPU                       | One RTX 5090; other users' GPU processes were not interrupted     |

### Limitations

Only one seed was run; cached visual embeddings were used; BF16 replaced any unspecified original numeric precision; and the exact paper code snapshot was unavailable. The public EEG cache was reused on the group server, while the repository documents an independent download path. The token and AI-time figures are transparent estimates and should be replaced with account-exported values if available before submission.

### Expected repository outputs

Runnable scripts, pinned dependencies, data-preparation instructions, resolved configurations, validation logic, aggregation code, compact JSON/CSV results, this report, HowTo.md, and a reusable SSH skill are all included. No passwords, API keys, private keys, or private server addresses are stored in the repository.

## 7. References

- [1] Zheng, J. et al. Learning Brain Representation with Hierarchical Visual Embeddings. ICLR 2026. <https://arxiv.org/abs/2602.07495>
- [2] Brain-HIVE source repository. <https://github.com/ssshamiii/Brain-HIVE>
- [3] Reproduction repository. <https://github.com/HUI-HUI-lab/brainhive-retrieval-reproduction>
- [4] THINGS-EEG data. <https://huggingface.co/datasets/Haitao999/things-eeg>
- [5] Cached visual embeddings. [https://huggingface.co/datasets/fakekungfu/Brain-HIVE\\_Visual\\_Embeddings](https://huggingface.co/datasets/fakekungfu/Brain-HIVE_Visual_Embeddings)